

Device Emulator Requirements

Minimum Viable Product Requirements

Functional Requirements

1. The emulator must be able to take user-specified device credentials, and transmit those to the registration endpoint.
 - The emulator should implement TLS for data transmission here.
2. The emulator must take a file path to a CSV file and transmit the contents of the CSV file to a user-specified endpoint.
 - The emulator should implement mTLS for data transmission here.
3. Dummy device certificate renewal
4. Dummy device certificate expiration
5. Dummy device new certificate creation
6. Saving the registration, credential, and certificate data of multiple dummy devices in persistent, secure **local** storage.

Non-Functional Requirements

7. The device emulator must run like any other command line tool in both Windows and Linux terminals.
8. The emulator architecture must be documented in a drawio UML diagram.
9. Complex or interesting workflows should have sequence diagrams
 - This includes workflows that should be tested to ensure the emulator itself is not failing
10. The emulator should have a proper “help” flag as well as a “manual” that will act as a guide for a user of the emulator. It should properly document the tools usage.
 - I recommend reading some manpages for tools like “grep” or “ls” to get the general idea of what this requirement is asking for.
11. Proper github practices followed
 - Feature branching
 - Semantic versioning
12. Create/add to Workflow file to integrate this tool into our CI/CD pipeline (Github Actions)

Enhancements

Functional Enhancements

1. Saving the registration, credential, and certificate data of multiple dummy devices in persistent, secure **local** storage.
2. Switch between saved dummy devices.
3. Send trail data as any one of the saved dummy devices, more than one device, or all of the dummy devices.
4. Perform benchmark tests for varying levels of dummy devices transmitting data, as well as how much data is being transmitted.

5. Testing data transmission when mTLS is not used.
6. Attempting DOS of server/load test
 - a. Could be fun, why not?
 - b. load tests are industry standard
 - c. if the devices all plan to be transmitting at the same time every day (Midnight) then we may want to test what happens when we scale to 100+ devices rather than 2-3. Latency on the device side won't be felt by the customer, but the consequences are unpredictable and that's not really something we want to leave alone.
7. Clear local storage of one, some, or all dummy device data

Non-Functional Requirements

8. Graphical User Interface for less technical users
9. CI/CD testing to ensure the emulator has no internal faults that could lead to false positives in the main repo's tests
10. Proper package releases
 - a. Installation scripts
 - b. READMEs
 - c. Binary distributions